

Synthesizing reduced form estimates from heterogeneous policy instruments: evidence from e-cigarette regulation

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- **Presenter's tobacco-related funding, past 10 years:** I have received no funding from tobacco or pharmaceutical companies in the previous 10 years.

What this talk is about

- **The question:** what happens to youth smoking when there is a policy-induced shock to youth vaping
- **The problem:** a dozen credible policy evaluations but no common scale to pool estimates
- **What we do:** standardize each policy by how much it moved youth vaping, then pool the estimates for its effect on youth smoking
- **What it adds:** a magnitude and a confidence interval, where prior syntheses could only report direction — and a generalizable framework

What we want to know: the causal effect of e-cigarette use on youth smoking

- On one hand, a large cohort literature follows individuals who vape and asks what they do later
- On the other, a growing set of policy evaluations takes a different approach — using policy changes that shift vaping for reasons unrelated to the individual
- These designs use policy changes to approximate, with real-world data, the experiment nobody can run: randomly assigning youth to e-cigarette access
- This study is about that second group — but every study evaluates a different policy, so we do not yet know how to combine them



Where the variation comes from

- States adopted age-of-sale laws, taxes and flavor restrictions at different times; disposable-vape bans are the newest example
- That staggered timing is what makes comparisons possible: states that adopted a policy versus states that had not yet, before and after
- What these designs give us: variation across states and over time, rather than across individuals who chose to vape
- Two things to keep in mind: the estimate is a net effect across all channels, and it relies on treated and comparison states trending together beforehand

Starting point for our study: Begh et al. (2025)

- The first systematic assessment of population-level studies

21

population-level studies on
e-cig availability → youth
smoking

15

found an inverse association

12

of those were significant

4

found a direct association

2

of those were significant

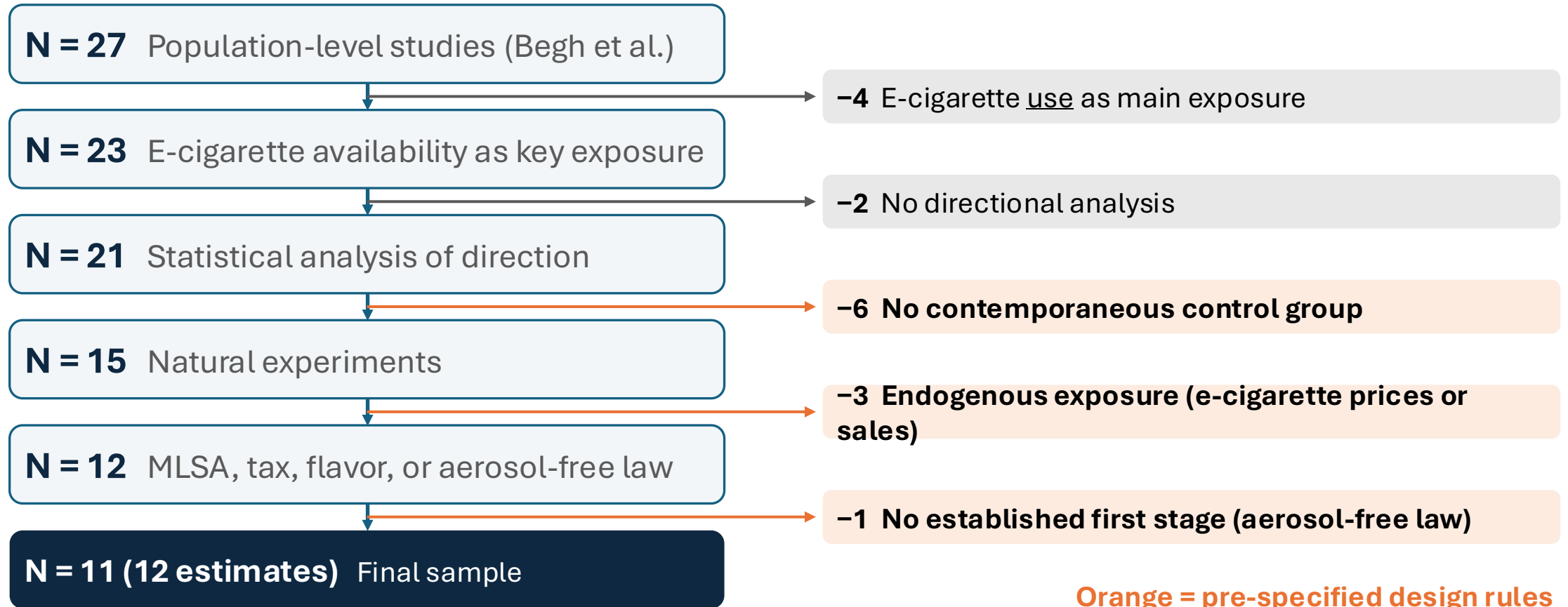
- Evidence was graded **very low certainty**, limited by risk of bias and inconsistency
- Summary of direction only, no pooled magnitude

The standardization problem — and what we do

- Why the review stopped at direction: the studies evaluate different policies, with no common treatment scale
- That is the standardization problem, and solving it is what this paper does
- We keep only the natural experiments, put their estimates on a common scale, and pool them
- Two questions: does the direction hold, and does precision improve on direction alone?
- The method is reusable: as new policies are evaluated, their effect on vaping can be translated into an implied effect on smoking

Study selection: 27 → 11 studies (12 estimates), rules fixed in advance

Goal: isolate the natural experiments whose policies actually change e-cigarette use.



The 11 studies at a glance: 7 MLSA, 3 tax, 1 flavor restriction



Data: YRBSS, MTF, NYTS, NSDUH, U.S. Natality files, CPS-TUS, MA-YHS

Populations: mostly high school-aged youth

Risk of bias (Begh et al. criteria)



Assessment considered 7 domains of bias:

- Confounding
- Selection of participants in the study
- Misclassification of exposure
- Deviations from intended exposure
- Outcome measurement
- Missing data
- Selection of reported results

Solving the standardization problem using a common dose

Policies aren't directly comparable (e.g., a \$1 tax versus a flavor restriction).

Measure every policy by its estimated effect on youth vaping, in \$1-tax-equivalent units.

Estimated change in youth vaping:

\$1 tax



Policy X



= "X units" of a \$1 tax

Dose of policy X = its effect on vaping ÷ a \$1 tax's effect on vaping

Step 1: how big a "dose" is each policy?

Estimate pooled effect on youth vaping using fixed-effects meta-analysis:

Policy	Estimates	Effect on youth vaping	I ²
Minimum legal sale age (MLSA)	k = 4	-19.1%	90.1%
\$1 e-cigarette tax	k = 4	-20.3%	41.9%
Flavor restriction	k = 1	-55.1%	single study

Express each policy in \$1-tax units

MLSA:

$$-19.1\% \div -20.3\% = 0.94$$

Flavor:

$$-55.1\% \div -20.3\% = 2.71$$

Dose in \$1-tax units

MLSA 0.94

\$1 tax 1.00

Flavor restriction 2.71

Step 2: putting one estimate on the common scale

Worked example: Friedman (2015), a study of state MLSA laws

Step	Operation	Result
As published	percentage-point effect	0.67 ppt
On the outcome scale	÷ outcome mean (10.3%)	6.5%
On the common scale	÷ MLSA dose (0.94)	+6.9%

Dose in \$1-tax units	
MLSA	0.94
\$1 tax	1.00
Flavor restriction	2.71

The standard error is divided by the same factor, so precision is treated consistently

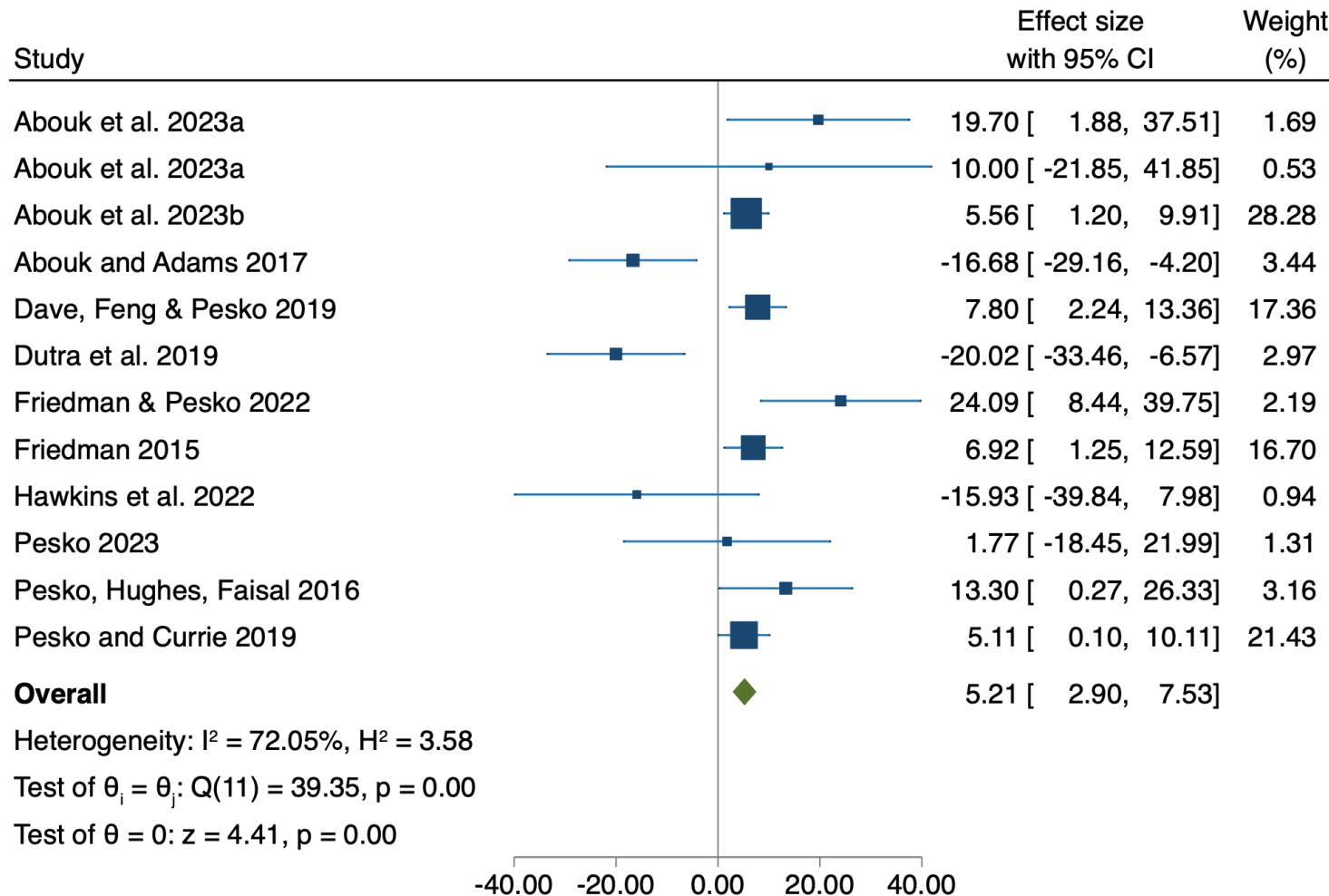
Repeat for all 12 estimates and they are finally on one common axis — full details of each step are in the paper

Friedman, A. S. (2015). How does electronic cigarette access affect adolescent smoking? *Journal of Health Economics*, 44, 300-308.

How to interpret the forest plot

- Each row is one study estimate, now expressed per \$1-tax-equivalent reduction in youth vaping
- The square is the point estimate; its size is the weight the study carries in the pooled result
- The horizontal line through it is the 95% confidence interval; the vertical line is zero
- Anything to the right of zero means restricting e-cigarette access is associated with more youth smoking
- Weights are inverse-variance: precision determines weight — not the size or the direction of the estimate
- The diamond at the bottom is the pooled estimate, and its width is that estimate's confidence interval

Pooled fixed effects: +5.2% in youth smoking (95% CI 2.9, 7.5)



The pooled estimate

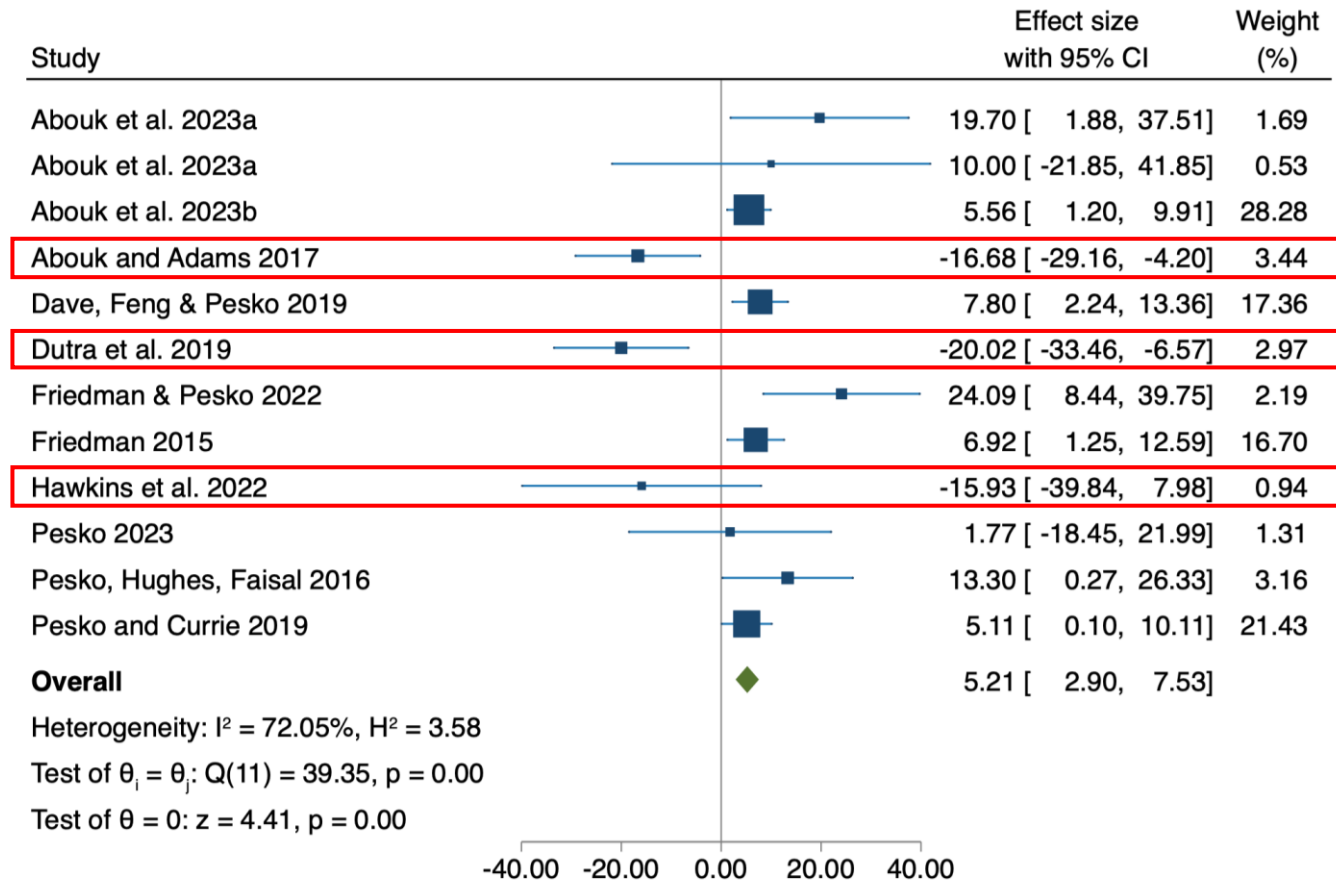
$I^2 = 72.1\%$

Four estimates carry more than 80% of the weight

Why fixed effects is primary: a common policy-intensity scale, similar designs, and broadly comparable populations and contexts

Not all studies point the same way

Panel A. Fixed effects model



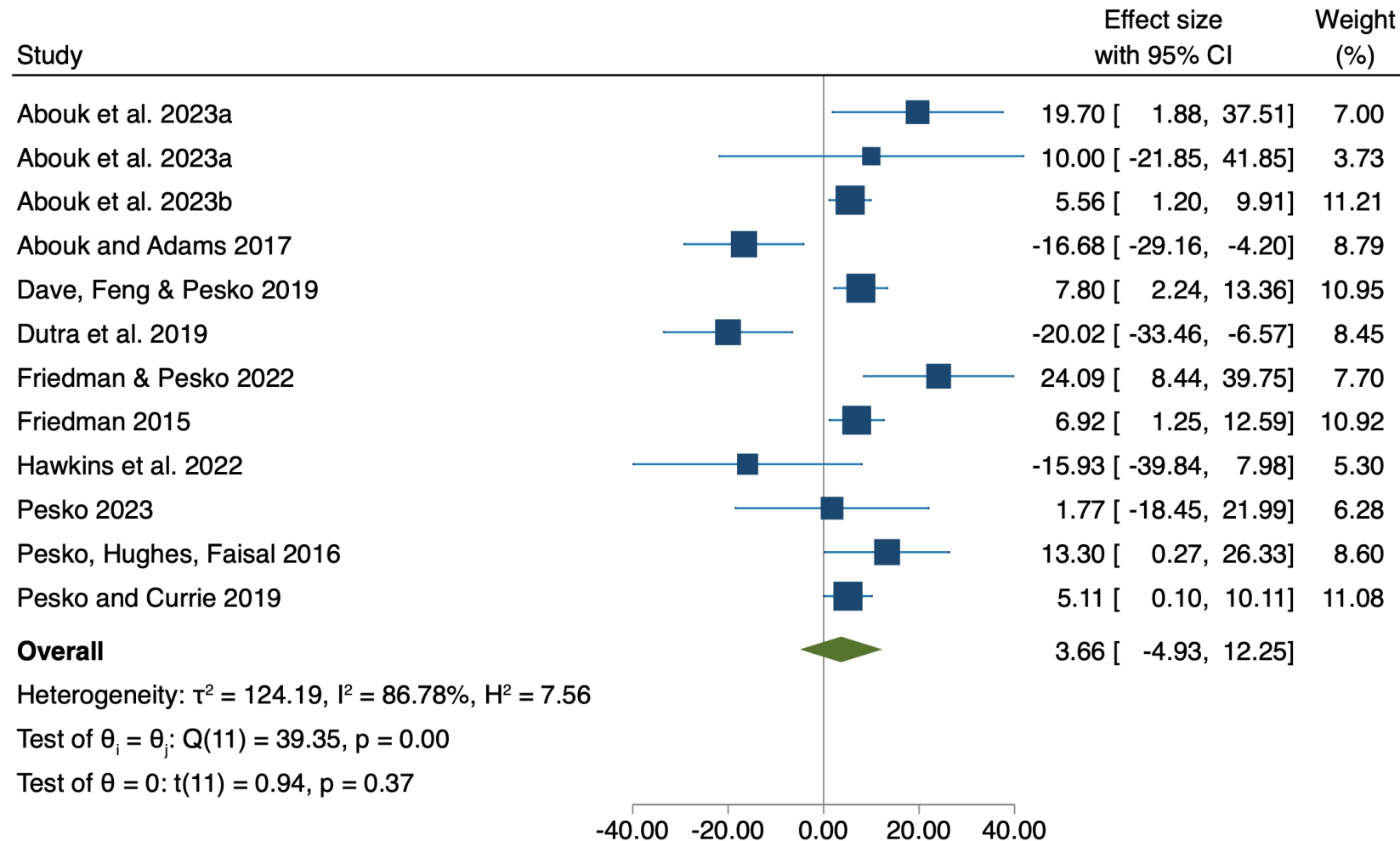
Three of the twelve estimates are negative

- Abouk & Adams 2017 (MLSA)
- Dutra et al. 2019 (MLSA)
- Hawkins et al. 2022 (flavor)

We pool all of them: nothing is dropped for pointing in the unexpected direction

Dispersion around a common effect is to be expected

Random effects: +3.7% (95% CI -4.9, 12.3)



The pooled estimate

Same direction, less
precise

$I^2 = 86.9\%$

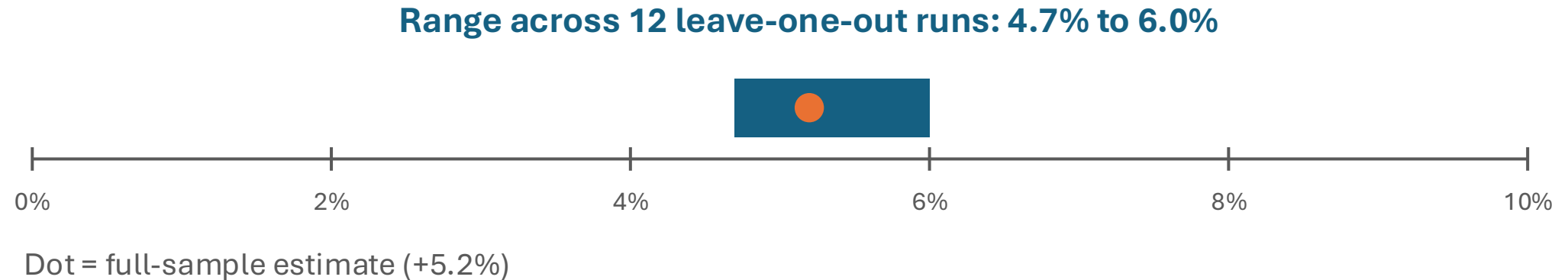
Random effects tests
sensitivity to weighting
assumptions, given limited
precedent for pooling across
policy instruments

What the weights do: fixed versus random effects

Study	Fixed effects	Random effects	Change
Abouk et al. 2023b — precise estimate	28.3%	11.2%	loses influence
Pesko 2023 — imprecise estimate	1.3%	6.3%	gains influence

- **Fixed effects:** weight is inverse-variance — the more precise the estimate, the more it counts
- **Random effects:** a common between-study variance is added to every study, pulling the weights toward equality
- **The two models differ mainly because influence shifts from precise studies to imprecise ones**

Leave-one-out: fixed effects stays between +4.7% and +6.0%



- ✓ Significant in every run
- ✓ Dropping Dutra (2019) or Abouk & Adams (2017) lowers I^2 (heterogeneity across studies) the most

Excluding the two critical-risk studies: the two models converge on +6.8%

	Fixed effects	Random effects	I ²
All 11 studies	+5.2% (2.9, 7.5)	+3.7% (−4.9, 12.3)	72.1%
Excluding 2 critical-risk studies*	+6.8% (4.4, 9.2)	+6.8% (3.6, 10.1)	26.7% (FE) / 0% (RE)

*The excluded studies are Dutra et al. (2019) and Abouk & Adams (2017) — the same two the leave-one-out analysis identified as the source of the heterogeneity.

- The estimate rises slightly but stays close to the leave-one-out range (4.7–6.0%), and the intervals overlap substantially.
- Fixed- and random-effects models now agree → the two highest risk studies were driving heterogeneity in pooled estimate

How this compares with the cohort literature

Cohort meta-analyses converge around **3x odds of later smoking** among non-smoking e-cigarette users (e.g., Kaur et al. 2026, Khouja et al. 2021)

Why the estimates differ

- A different question: individual choices to vape, versus policy-driven changes in vaping
- They condition on never-smokers, which rules out substitution by construction
- Common liabilities push those estimates away from the null

What each is good for

- Cohort studies describe individual trajectories — who transitions between products
- Natural experiments give the population-level net effect

Takeaways: e-cigarettes

Restricting youth e-cigarette access is associated with more youth smoking

+5.2% (95% CI 2.9, 7.5) per \$1-tax-equivalent reduction in youth vaping

Consistent in direction across models,
leave-one-out and policy type

Strongest (+6.8%, both models) after
dropping the critical-risk studies

What has to hold, and what comes next

What has to hold:

1. E-cigarette policies affect smoking mainly through vaping
2. Effects scale roughly linearly with the size of the vaping reduction
3. Each underlying DiD design satisfies parallel trends — 9 of 11 assess them using event studies, pre-period leads, or falsification

Limitations, and what comes next:

- No established risk-of-bias tool for population-level quasi-experiments
- The flavor restriction first stage rests on a single study
- A high-weight study uses rural pregnant teens (robust to dropping it)
- Youth only
- **The search is bounded by the April 2023 review cutoff — we are seeking funding to expand the search and update the evidence base**

Takeaways: methods

A general tool for synthesizing policies in any domain that share an intermediate outcome but lack a common treatment scale

One estimate that uses all available studies

Reusable as new policies are evaluated

Thank you

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Working paper available on SSRN (<https://ssrn.com/abstract=7500979>)